



NAUVAL ZULFIKAR

Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Cambridge, 2026-10-01) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264
[LinkedIn](#) · [GitHub](#) · [Portfolio](#)
Visa status: UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate with a First Class from Aston University, specialising in blockchain-integrated LLM analysis for supply-chain customer reviews. Extensive experience in applied ML across banking, government, and start-up environments. Eager to contribute to research on LLM safety, focusing on inference-time safety monitoring and mechanistic interpretability.

EDUCATION

Aston University, Birmingham, UK

MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan

BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- **Distinctions:** Business Data Analysis, Consumer Behaviour, Marketing Research

INFORMAL EDUCATION

Indonesia AI

NLP Engineer Training · EQF Level 5 · May – Nov 2023

Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project.

Distinctions: Python Programming, Text Summarisation, Voice Recognition.

Qarir Generator (Dubai, UAE)

Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023

Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project.

Distinctions: Python, SQL, Pandas/NumPy.

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini, augmented with retrieval over a curated behavioural-literature knowledge base.
- github.com/nauvalZulfikar/LLM-Shipper-Profiles

MSc Dissertation Pipeline (DeBERTa-v3 fine-tune) (2024)

- Fine-tuned DeBERTa-v3 on multi-tier supplier review data. Evaluated emergent biases against literature baselines.
- Integrated into a larger decision system. Graded 1:1.

Crowd-Shipping ABM Behavioural Sweep (2026)

- Conducted a 450-run parameter sweep with auto-generated findings linking results to literature.
- github.com/nauvalZulfikar/Crowd-Shipping-ABM-

VRPOD MILP Optimisation (2026)

- Replicated Archetti et al. (2016) VRPOD problem using `scipy.optimize.milp`, achieving a 38.5% cost reduction.
- github.com/nauvalZulfikar/VRP-MILP

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung

Data Scientist & Project Manager · Sep 2022 – Present

Automated ETL pipelines and built self-service dashboards, achieving **45% faster project delivery** and **20% data-efficiency gain**.

Syncwell, Birmingham, UK

Marketing Data Scientist · Jul – Sep 2024

Implemented predictive ML and personalisation strategies, resulting in a **40% CTR increase** and **22% follower growth**.

PCOS Challenge, Birmingham, UK

Strategic Data Scientist Specialist · Jun – Aug 2024

Developed a customer attribution model, reducing manual workload by **80%**.

Bank Muamalat Indonesia, Jakarta

Lead Data Scientist · Nov 2019 – Aug 2022

Developed NLP models for credit portfolio analysis, achieving a **5.3% investment return**.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Comparative evaluation of multiple ML regressors for retail sales forecasting accuracy.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API + RAG, sentence-transformers, classification, regression, time-series forecasting.

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit.

Mathematical/Formal: MILP (Pyomo, `scipy.optimize.milp`, HiGHS), Agent-Based Modelling (mesa), Discrete-Event Simulation (simpy), Bayesian methods.

Governance: Model validation, model reliability, audit-grade documentation, AWS, Docker, Git, ETL/orchestration.

LANGUAGES

Indonesian (Native) · **English** (Fluent, IELTS 7.0 overall) · **Japanese** (Intermediate)

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

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Dr. Viktor Pekar

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